



## GENSET NO LOAD & LOAD TEST REPORT

|               |                      |                |        |
|---------------|----------------------|----------------|--------|
| Date :        | 06 / FEBRUARY / 2016 |                |        |
| PI Model :    | PI 563P              | Type :         | CLOSE  |
| Prime Power : | 500 KVA              | Canopy Color : | ORANGE |
| Frq. :        | 60 Hz                | P.F            | 1      |
| Serial No :   |                      |                |        |
| Job No :      | 7506                 |                |        |

CLIENT : KSA STOCK

PROJECT :

### GENSET DETAILS :

|                  |          |            |                 |             |                   |
|------------------|----------|------------|-----------------|-------------|-------------------|
| Engine :         | PERKINS  | Model No : | 2506A – E15TAG2 | Serial No : | MGBF5120 N 00924A |
| Alternator :     | STAMFORD | Model No : | HCI544D1        | Serial No : | A14K487202        |
| Controller :     | COMAP    | Model No : | AMF9            | Serial No : | 15014ECC          |
| C. Breaker :     | METASOL  | Amps       | 800 A           | Model No :  | ABN 803C          |
| C. Transformer : | ZIEGLER  | Amps       | 1000/5 A        | Model No :  | N/A               |
| AVR :            | STAMFORD | Model No : | STAMFORD        | Serial No : | 104277 – 1701     |

### GENSET RATING :

|                             |          |                   |          |
|-----------------------------|----------|-------------------|----------|
| Testing Full Load Current : | 577 Amps | Nominal Current : | 722 Amps |
|-----------------------------|----------|-------------------|----------|

### LOAD TEST WILL BE IMPLEMENTED AT AMBIENT CONDITIONS

|               |       |            |  |
|---------------|-------|------------|--|
| Temperature : | 26° C | Altitude : |  |
|---------------|-------|------------|--|

### LOAD TEST

| LOAD | RUN TIME | FREQ | SPEED | VOLTAGE L-L |     |     | CURRENT |     |     | POWER | COOLANT TEMP. | OIL PRESSURE | CHARGING ALTERNATOR |
|------|----------|------|-------|-------------|-----|-----|---------|-----|-----|-------|---------------|--------------|---------------------|
|      |          |      |       | L1          | L2  | L3  | L1      | L2  | L3  |       |               |              |                     |
| %    | Min      | Hz   | RPM   | V           | V   | V   | A       | A   | A   | Kw    | °C            | BAR          | VDC                 |
| 0%   | 5        | 60.0 | 1800  | 380         | 380 | 380 | --      | --  | --  | --    | 74            | 4.76         | 28.0                |
| 26%  | 5        | 60.0 | 1800  | 380         | 380 | 378 | 147     | 150 | 147 | 98    | 88            | 4.60         | 28.3                |
| 54%  | 5        | 60.0 | 1800  | 380         | 381 | 379 | 308     | 310 | 308 | 203   | 89            | 4.40         | 28.2                |
| 76%  | 5        | 60.0 | 1800  | 380         | 381 | 380 | 433     | 436 | 433 | 287   | 89            | 4.32         | 28.2                |
| 101% | 10       | 60.0 | 1800  | 380         | 381 | 380 | 578     | 581 | 577 | 382   | 91            | 4.20         | 28.2                |
| 111% | 10       | 60.0 | 1800  | 380         | 381 | 380 | 633     | 633 | 631 | 417   | 94            | 4.16         | 28.2                |

### LOAD ACCEPTANCE TEST:

| SUDDENLY BLOCK LOAD | VOLTAGE   | FREQUENCY   | CURRENT | RESPONSE TIME |
|---------------------|-----------|-------------|---------|---------------|
| %                   | V         | Hz          | A       | SEC           |
| 53%                 | 380       | 60.0        | 308     | 0             |
| 0%                  | 382 – 380 | 61.4 – 60.0 | 0       | 5             |
| 53%                 | 378 – 380 | 59.6 – 60.0 | 308     | 5             |

### PROTECTION SYSTEM TEST:

| OVER FREQUENCY SD | UNDER FREQUENCY SD | OVER VOLTAGE SD | UNDER VOLTAGE SD | OIL PRESSURE WRN/SD | COOLANT LEVEL SD | COOLANT TEMP WRN/SD |
|-------------------|--------------------|-----------------|------------------|---------------------|------------------|---------------------|
| 110%              | 85%                | 460             | 350              | 2.0 / 1.0           | OK               | 95 / 105            |

### ACCESSORIES DETAILS :

|  |       |  |           |  |
|--|-------|--|-----------|--|
|  | Model |  | Serial No |  |
|  | Model |  | Serial No |  |
|  | Model |  | Serial No |  |
|  | Model |  | Serial No |  |

### Test Carried out by PI Supervisor :

|             |                      |                                    |        |
|-------------|----------------------|------------------------------------|--------|
| Name :      | Eng. Malek Yousef    | Witnessed and approved by Client : | Name : |
| Signature : |                      | Signature :                        |        |
| Date :      | 06 / FEBRUARY / 2016 | Date :                             |        |



## GENSET PRE-TEST CHECKING

|               |                      |                |        |
|---------------|----------------------|----------------|--------|
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| Controller : | COMAP    | Model No : | AMF9            | Serial No : | 15014ECC          |

### SAFETY CHECKS

- Set clean and fully assembled ☒
- No loose materials ☒
- Electrical bonding complete ☒
- Pipe work & cables are secure with no trip hazards ☒

### COOLING SYSTEM

- Radiator clean, free from obstructions ☒
- Radiator filled with correct coolant & cap replaced ☒
- Radiator, engine & pipe work checked for leaks ☒
- Belts checked for alignment, tension and damage ☒
- Guards in place and secure ☒

### FUEL SYSTEM

- Isolating and solenoid valves checked ☒
- Engine return open ☒
- Tank fuel gauge ☒
- Transfer pump functions tested ☒

### LUBRICATION SYSTEM

- Engine oil pan filled to correct level ☒
- Level alarms checked ☒
- Drain valves/pipes checked for leaks ☒

### STARTING SYSTEM

- Batteries installed with correct capacity ☒
- Batteries, cables are secured ☒
- Battery charger commissioned ☒

### EXHAUST SYSTEM

- Check security of bellows, pipe work & muffler ☒
- Check all flanges, joints & welds ☒
- Check tail pipe and rain cap are clear ☒
- Pre-lubricate turbo charger if required ☒

### ELECTRICAL SYSTEM

- Visual check completed ☒
- Check functionality ☒
- Check software versions and update as required ☒
- Select the parameters on system & GENSET ☒
- Check GENSET control signals and remote control ☒
- Check the emergency stop controls ☒
- Verify engine & alternator protection settings ☒
- Check cable installed correctly & torque marked ☒
- Verify cable flexibility at generator set ☒
- Test certificates available for all cables ☒
- Check supply protection ☒
- Check auxiliary supplies voltage & phase ☒

Comments.....

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### Test Carried out by PI Supervisor :

Name : Eng. Malek Yousef  
Signature :  
Date : 06 / FEBRUARY / 2016

### Witnessed and approved by Client :

Name :  
Signature :  
Date :